# Quadratic equations

**1** The equation $x\left(x-7\right)=-10$ has a positive integer solution of $x=5$. Find its other solution.

**2** Find the value of $a$ if:

**a** The nonzero solution of the equation $x\left(x+a\right)=0$ is $x=10$.

**b** The nonzero solution of the equation $x\left(x+a\right)=0$ is $x=-1$.

**3** Solve the following equations:

**a** $m^2=14m$

**b** $m^2=m+20$

**c** $m^2=7m-6$

**d** $m^2-11m=-30$

**e** $m^2-27m=-182$

**f** $\frac{m}{6}\left(m-10\right)=0$

**g** $-3m\left(m+2\right)=0$

**h** $3\left(n^2+8\right)=-18n$

**i** $2y-6y^2=0$

**j** $3y-15y^2=0$

**k** $\left(y-3\right)^2=2y+2$

**l** $\left(y+1\right)^2=4y+4$

**m** $15-11b-12b^2=0$

**n** $6k^2=-7+13k$

**4** Solve the following equations:

**a** $x^2-64=0$

**b** $x^2+12x=0$

**c** $x^2+13x+42=0$

**d** $x^2+4x-21=0$

**e** $x^2-14x+40=0$

**f** $x^2-5x-50=0$

**g** $x^2-22x+121=0$

**h** $-x^2+x+6=0$

**i** $x\left(x+18\right)=-80$

**j** $x\left(x+2\right)=48$

**k** $x^2+8x+16=36$

**l** $x^2=13x+114$

**m** $3x^2-7x-20=0$

**n** $5x^2+22x+8=0$

**o** $-4x^2+25x-36=0$

**p** $-5x^2=-53x+72$

**q** $-10x+x^2=2-7x-x^2$

**r** $-8x+x^2=-6-x-x^2$

**s** $6x^2-27x+28=-3x^2+3x+3$

**t** $\left(5x^2-13x+6\right)\left(2x^2-13x+20\right)=0$

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**5** Solve the following equations:

**a** $\dfrac{x^2+3x}{4}=7$

**b** $\dfrac{x^2-5x}{8}=3$

**c** $x-\dfrac{28}{x}=3$

**d** $x-\dfrac{45}{x}=4$

**e** $\dfrac{9+3x}{2x}=x$

**f** $\dfrac{2x^2-19x}{3}=20$

**g** $\dfrac{20}{x}-3x=-11$

**h** $-\dfrac{6}{x}+\dfrac{2x}{7}=-\dfrac{8}{7}$

**i** $\dfrac{x+1}{2}-\dfrac{x+2}{3}=\dfrac{1-x}{3x+1}$

**j** $\dfrac{7}{x\left(x+3\right)}+5=\dfrac{x+4}{x}$

## Applications

**6** On the graph of $y=x^2-4$ , there are two points where $y=12$. Find the $x$-coordinates of these two points.

**7** The $x$-intercepts of a graph occur when $y=0$. Find the $x$-intercepts of the following graphs:

**a** A parabola with equation $y=x^2+x-6$

**b** A circle with equation $\left(x+2\right)^2+\left(y-4\right)^2=41$

**8** Software engineers are designing a self-serve checkout system for a supermarket. They notice that the traffic through the store during the day is described by the function:

$$C=-t\left(t-12\right)$$

where $C$ is the number of customers and $t$ is the number of hours after the store opens.

To meet the peak demand, the engineers allow for an extra checkout machine to automatically turn on when the number of customers first reaches $27$ people, and to automatically turn off when it next falls below $27$ people.

**a** Find the times $t$ when the number of customers is equal to $27$ people.

**b** Hence, after how many hours will the extra checkout machine turn on after opening?

**c** How many hours will the extra machine be on for?

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